Table S1. Characteristic of studies included for vaccine effectiveness

First author	Published time	Study design	Location	Study group	Vaccine type	Vaccination status	Variants of SARS-CoV-2	Vaccinated people with SARS-CoV-2 infection (n/N)	Unvaccinated people with SARS-CoV-2 infection (n/N)	Vaccine effectiveness (%)	Vaccinated people with symptomatic COVID-19 (n/N)	Unvaccinated people with symptomatic COVID-19 (n/N)	Vaccine effectiveness (%)	Vaccinated people with COVID-19 hospital admissions (n/N)	Unvaccinated people with COVID-19 hospital admissions (n/N)	Vaccine effectiveness (%)	Vaccinated people with COVID-19 ICU admissions (n/N)	Unvaccinated people with COVID-19 ICU admissions (n/N)	Vaccine effectiveness (%)	Vaccinated people with COVID-19-related death (n/N)	Unvaccinated people with COVID-19-related death (n/N)	Vaccine effectiveness (%)	risk of bias
Laith J. Abu-Raddad	2021.7	case control	Qatar	general population	BNT162b2	1	Alpha (B.1.1.7)	892/2133	18075/35801	30 (23, 36)	30/91	468/905	54 (26, 72)	..	..	..	..	..	..	..	..	..	low
							Beta (B.1.351)	1329/2909	20177/40103	17 (10, 23)	45/80	348/706	0 (0, 19)	..	..	..	..	..	..	..	..	..	
							any	..	..	..	139/359	1966/3851	39 (24, 52)	..	..	..	..	..	..	..	..	..	
						2(⩾14)	Alpha (B.1.1.7)	50/515	16354/32293	90 (86, 92)	0/20	401/782	100 (82, 100)	..	..	..	..	..	..	..	..	..	
							Beta (B.1.351)	179/877	19396/38273	75 (71, 79)	0/14	300/586	97 (92, 100)	..	..	..	..	..	..	..	..	..	
							any	..	..	..	3/112	1692/3278	97 (92, 100)	..	..	..	..	..	..	..	..	..	
Yoel Angel*	2021.5	cohort study	Israel	Health Care Workers	BNT162b2	1(7-28)		37/117389	8/15091	36 (-51, 69)	31/17420	37/15128	89 (83, 94)	..	..	..	..	..	..	..	..	..	low
						2(⩾7)		19/168571	17/25359	86 (69, 93)	8/168571	38/25359	97 (94, 99)	..	..	..	..	..	..	..	..	..	
						2(⩾21)		4/95689	11/15726	94 (78, 98)	2/95691	23/15749	99 (94, 100)	..	..	..	..	..	..	..	..	..	
Tariq Azamgarhi	2021.6	cohort study	UK	Health Care Workers	BNT162b2	1(<14)		16/1408	13/825	20 (-147, 69)	..	..	..	..	..	..	..	..	..	..	..	..	low
					BNT162b2	1(⩾14)		7/1392	13/812	70 (6, 91)	..	..	..	..	..	..	..	..	..	..	..	..	
Zacay, G.*	2021.7	cohort study	Israel	general	BNT162b2	1(⩾14)		21/100000	53/100000	61 (49, 71)	..	..	..	..	..	..	..	..	..	..	..	..	low
						2(0-6)		9/100000	53/100000	82 (71, 89)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(⩾7)		6/100000	53/100000	89 (82, 94)	..	..	..	..	..	..	..	..	..	..	..	..	
Williams, C.	2021.7	cohort study	Canada	staff in long term care	BNT162b2	2(⩾14)	Gamma (P.1)	4/43	11/40	62 (2, 88)	4/43	5/40	26 (-158, 79)	..	..	..	..	..	..	..	..	..	moderate
				elder residents in long term care	mRNA-1273	2(⩾14)	Gamma (P.1)	19/48	10/12	53 (27, 70)	11/48	8/12	66 (34, 82)	..	..	..	6/47	7/12	79 (48, 91)	..	..	..	
Francesco Paolo Bianchi	2021.5	cohort study	Italy	Health Care Workers	BNT162b2	1(14-20)		..	..	62 (19, 82)	..	..	..	..	..	..	..	..	..	..	..	..	moderate
						1(21-27)		..	..	88 (52, 97)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(⩾7)		..	..	96 (82, 99)	..	..	..	..	..	..	..	..	..	..	..	..	
Amadea Britton*	2021.3	cohort study	USA	nursing home residents	BNT162b2	1(⩾14)		25/4172	39/3612	63 (33, 79)	..	..	..	..	..	..	..	..	..	..	..	..	moderate
Martínez-Baz, I.	2021.5	cohort study	Spain	adults who had been close contacts of laboratory-confirmed COVID-19 cases from January to April 2021.	Mix	1(⩾14)		198/869	6980/19580	35 (25, 44)	135/869	5306/19580	42 (31, 52)	10/869	548/19580	72 (47, 85)	..	..	..	..	..	..	low
						2(⩾14)		62/512	6980/19580	66 (57, 74)	26/512	5306/19580	82 (74, 88)	1/512	548/19580	95 (62, 99)	..	..	..	..	..	..	
					AZD1222	1(⩾14)		99/524	6980/19580	..	67/524	5306/19580	..	1/524	548/19580	..	..	..	..	..	..	..	
					BNT162b2	2(⩾14)		61/492	6980/19580	..	25/491	5306/19580	..	1/491	548/19580	..	..	..	..	..	..	..	
					mRNA-1273	1(⩾14)		90/310	6980/19580	..	62/310	5306/19580	..	7/310	548/19580	..	..	..	..	..	..	..	
						1(⩾14)		9/35	6980/19580	..	6/35	5306/19580	..	2/35	548/19580	..	..	..	..	..	..	..	
						2(⩾14)		1月21日	6980/19580	..	1/21	5306/19580	..	0/21	548/19580	..	..	..	..	..	..	..	
Swift, M. D.	2021.4	cohort study	USA	healthcare personnel	BNT162b2	1(>14) and 2(0-14)		..	..	78 (77, 82)	..	..	..	..	..	..	..	..	..	..	..	..	low
						2(>14)		..	..	97 (95, 98)	..	..	..	..	..	..	..	..	..	..	..	..	
					mRNA-1273	2(>14)		..	..	99 (90, 100)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(>14) and 2(0-14)		..	..	91 (81, 96)	..	..	..	..	..	..	..	..	..	..	..	..	
					mRNA-1273+BNT162b2	1(>14) and 2(0-14)		98/3204	997/23931	..	66/3204	876/23931	..	..	..	..	..	..	..	..	..	..	
						2(>14)		30/44002	997/23931	..	22/44002	876/23931	..	..	..	..	..	..	..	..	..	..	
Mazagatos, C.	2021.6	case-control	Spain	elderly long-term care facility residents	mRNA-1273+BNT162b2	1(⩾14)		1559/..	6820/..	51 (37, 61)	..	..	..	404/..	2105/..	53 (26, 70)	..	..	..	..	..	56 (27, 73)	low
						2(0-6 for Comirnaty or 0-14 for Moderna)		413/..	7966/..	79 (67, 86)	101/..	2408/..	83 (61, 93)	..	..	..	..	..	..	..	..	96 (83, 99)	
						2(⩾7 for Comirnaty or ⩾14 for Moderna)		292/300425	8087/46099	71 (56, 82)	49/300182	160/38172	88 (75, 95)	..	..	..	..	..	..	..	..	97 (92, 99)	
Alyson M. Cavanaugh	2021.4	cross-sectional	USA	nursing home residents	BNT162b2	2(⩾14)		18/71	6/8	66 (41, 81)	6/71	5/8	87 (66, 95)	2/71	4/8	94 (74, 99)	..	..	..	1/71	2/8	94 (45, 99)	moderate
				Health Care Workers	BNT162b2	2(⩾14)		4/56	16/54	76 (33, 91)	2/56	15/54	87 (46, 97)	..	..	..	..	..	..	..	..	..	
Tenforde, M. W.	2021.5	case-control	USA	Hospitalized Adults Aged ⩾65 Years	mRNA-1273+BNT162b2	1(0-13)		22/49	146/287	3 (-94, 51)	..	..	..	..	..	..	..	..	..	..	..	..	moderate
						1(⩾14) and 2(0-13)		18/62	146/287	64 (28, 82)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(⩾14)		1/19	146/287	94 (49, 99)	..	..	..	..	..	..	..	..	..	..	..	..	
Hiam Chemaitelly	2021.7	case control	Qatar	general population	mRNA-1273	1(<7)	Alpha (B.1.1.7)	166/336	23661/47318	2 (0, 22)	..	..	..	..	..	..	..	..	..	..	..	..	low
							Beta (B.1.351)	535/1093	47359/94695	4 (0, 15)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	..	..	..	53/118	4012/8012	19 (0, 45)	..	..	..	..	..	..	..	..	..	
						1(7-13)	Alpha (B.1.1.7)	202/389	23681/47377	0 (0, 12)	..	..	..	..	..	..	..	..	..	..	..	..	
							Beta (B.1.351)	854/1420	47372/95032	0 (0, 0)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	..	..	..	85/153	4016/8049	0 (0, 10)	..	..	..	..	..	..	..	..	..	
						1(14-20)	Alpha (B.1.1.7)	32/205	23669/47197	82 (73, 88)	..	..	..	..	..	..	..	..	..	..	..	..	
							Beta (B.1.351)	270/786	47329/94412	48 (40, 55)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	..	..	..	18/78	4007/7972	70 (49, 84)	..	..	..	..	..	..	..	..	..	
						1(21-27)	Alpha (B.1.1.7)	9/169	23656/47161	94 (89, 98)	..	..	..	..	..	..	..	..	..	..	..	..	
							Beta (B.1.351)	114/545	47249/94181	74 (68, 79)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	..	..	..	4/54	3997/7948	92 (78, 98)	..	..	..	..	..	..	..	..	..	
						2(<7)	Alpha (B.1.1.7)	4/200	21454/42716	98 (95, 100)	..	..	..	..	..	..	..	..	..	..	..	..	

						Beta (B.1.351)	42/761	45280/89883	94 (92, 96)	..	..	..	..	..	..	..	..	..	..	..		
						any	..	..	..	0/62	3432/6802	100 (94, 100)	..	..	..	..	..	..	..	..		
					2(7-13)	Alpha (B.1.1.7)	1/120	21377/42636	99 (95, 100)	..	..	..	..	..	..	..	..	..	..	..		
						Beta (B.1.351)	18/516	45064/89648	96 (94, 98)	..	..	..	..	..	..	..	..	..	..	..		
						any	..	..	..	0/29	3399/6769	100 (87, 100)	..	..	..	..	..	..	..	..		
Pawlowski, C.*	2021.7	cohort study	USA	general population	BNT162b2	1(≥7)		401/2942986	1232/2851069	69 (65, 72)	..	..	..	..	..	..	..	..	..	..	..		low
						1(≥14)		211/2595549	960/2506231	79 (75, 82)	..	..	..	..	..	..	..	..	..	..	..		
						1(≥7 before 2)		293/766978	534/761511	46 (37, 53)	..	..	..	..	..	..	..	..	..	..	..		
						1(≥14 before 2)		103/419541	262/416673	61 (51, 69)	..	..	..	..	..	..	..	..	..	..	..		
						2(≥7)		82/1914500	563/1828466	86 (82, 89)	..	..	..	7/1915615	60/1837267	89 (76, 96)	0/1915733	9/1838072	100 (51, 100)	..	..	..	
						2(≥14)		59/1670896	468/1591900	88 (84, 91)	..	..	..	6/1671628	49/1599076	88 (73, 96)	0/1671730	6/1599732	100 (19, 100)	..	..	..	
					mRNA-1273	1(≥7)		97/946890	303/927716	69 (61, 75)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥14)		49/833098	241/814413	80 (73, 86)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥7 before 2)		88/369195	181/366881	52 (37, 63)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥14 before 2)		40/255403	119/253578	67 (52, 77)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(≥7)		7/495550	101/478322	93 (86, 97)	..	..	..	9/948311	63/932315	86 (72, 94)	0/495630	8/479979	100 (43, 100)	..	..	..	
						2(≥14)		6/417664	75/402416	92 (82, 97)	..	..	..	5/833681	52/817970	91 (76, 97)	0/417743	6/403589	100 (18, 100)	..	..	..	
Gabriel Chodick	2021.6	cohort study	Israel	general population	BNT162b2	1(13-24)		614/..	2484/..	51 (16, 72)	406/..	1692/..	54 (21, 74)	..	..	..	..	..	..	..	..	..	low
Gabriel Chodick	2021.5	cohort study	Israel	general population	BNT162b2	2(7-27)		728/27456	4514/60913	90 (79, 95)	..	..	..	..	..	..	..	..	..	..	..	..	low
Pilishvili, T.	2021.5	case control	USA	Health Care Personnel	mRNA-1273+BNT162b2	1(>14) and 2(0-6)		64/305	559/1538	82 (74, 87)	..	..	..	..	..	..	..	..	..	..	..	..	low
						2(≥7)		40/418	583/1425	94 (87, 97)	..	..	..	..	..	..	..	..	..	..	..	..	
Vasileiou, E.	2021.4	cohort study	Scotland		BNT162b2	1(0–6)		..	..	..	..	..	..	156/19917	7698/831226	75 (71, 79)	..	..	..	..	..	..	low
						1(7–13)		..	..	..	..	..	..	258/19690	7698/831226	54 (48, 59)	..	..	..	..	..	..	
						1(14–20)		..	..	..	..	..	..	154/13426	7698/831226	67 (61, 72)	..	..	..	..	..	..	
						1(21–27)		..	..	..	..	..	..	72/9119	7698/831226	76 (70, 81)	..	..	..	..	..	..	
						1(28–34)		..	..	..	..	..	..	30/6252	7698/831226	89 (83, 92)	..	..	..	..	..	..	
						1(35–41)		..	..	..	..	..	..	21/3822	7698/831226	78 (68, 85)	..	..	..	..	..	..	
						1(≥42)		..	..	..	..	..	..	32/6047	7698/831226	74 (65, 81)	..	..	..	..	..	..	
					AZD1222	1(0–6)		..	..	..	..	..	..	122/9222	7252/743142	72 (66, 77)	..	..	..	..	..	..	
						1(7–13)		..	..	..	..	..	..	139/8699	7252/743142	68 (61, 73)	..	..	..	..	..	..	
						1(14–20)		..	..	..	..	..	..	83/5742	7252/743142	73 (66, 79)	..	..	..	..	..	..	
						1(21–27)		..	..	..	..	..	..	34/2447	7252/743142	81 (72, 87)	..	..	..	..	..	..	
						1(28–34)		..	..	..	..	..	..	11/1666	7252/743142	88 (75, 94)	..	..	..	..	..	..	
						1(35–41)		..	..	..	..	..	..	⩽5/530	7252/743142	97 (63, 100)	..	..	..	..	..	..	
						1(≥42)		..	..	..	..	..	..	⩽5/51	7252/743142	59 (-296, 96)	..	..	..	..	..	..	
Noa Dagan	2021.2	cohort study	Israel	general population	BNT162b2	1(14-20)		..	..	46 (40, 51)	..	..	57 (50, 63)	..	..	74 (56, 86)	..	..	62 (39, 80)	..	..	..	low
						2(≥7)		..	..	92 (88, 95)	..	..	94 (87, 98)	..	..	87 (55, 100)	..	..	92 (75, 100)	..	..	..	
Maria Elena Flacco	2021.6	cohort study	Italy	general population	BNT162b2	1		..	..	55 (40, 66)	..	..	..	..	..	..	..	..	..	..	..	..	low
						2		..	..	98 (96, 99)	..	..	99 (96, 100)	..	..	..	..	..	..	..	..	98 (87, 100)	
						⩾1		..	..	95 (93, 96)	..	..	95 (92, 97)	..	..	..	..	..	..	..	..	97 (91, 99)	
					mRNA-1273	1		..	..	93 (74, 98)	..	..	..	..	..	..	..	..	..	..	..	..	
Thompson, M. G.	2021.7	cohort study	USA	health care personnel, first responders, and other essential and frontline workers	BNT162b2	1(≥14) and 2(0-13)		8/2005	156/3964	80 (60, 90)	..	..	..	..	..	..	..	..	..	..	..	..	low
						2(≥14)		3/1731	156/3964	93 (78, 98)	..	..	..	..	..	..	..	..	..	..	..	..	
					mRNA-1273	1(≥14) and 2(0-13)		3/982	156/3964	83 (40, 95)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(≥14)		2/770	156/3964	82 (20, 96)	..	..	..	..	..	..	..	..	..	..	..	..	
Eric J Haas	2021.5	cohort study	Israel	general population	BNT162b2	2(≥7)		3·1/100000	91·5/100000	95 (95, 96)	0·8/100000	32·5/100000	97 (97, 97)	0·3/100000	4·6/100000	97 (97, 98)	0·2/100000	2·7/100000	98 (97, 98)	0·1/100000	0·6/100000	97 (96, 97)	low
						2(≥14)		2·1/100000	91·5/100000	97 (96, 97)	0·6/100000	32·5/100000	98 (98, 98)	0·2/100000	4·6/100000	98 (98, 98)	0·1/100000	2·7/100000	98 (98, 99)	<0·1/100000	0·6/100000	98 (98, 98)	
Victoria Jane Hall*	2021.5	cohort study	UK	Health Care Workers	BNT162b2+AZD1222	1(>21)		71/87278	977/710587	70 (55, 85)	..	..	..	..	..	..	..	..	..	..	..	..	low
						2(≥7)		9/20978	977/710587	85 (74, 96)	..	..	..	..	..	..	..	..	..	..	..	..	
Catherine Hyams	2021.6	case control	UK	people aged at least 80 years	AZD1222	1		..	..	..	9/62	27/64	80 (36, 95)	..	..	..	..	..	..	..	..	..	low
					BNT162b2	1		..	..	..	18/108	117/296	71 (43, 86)	..	..	..	..	..	..	..	..	..	
Sheikh, A.	2021.6	cohort study	Scotland		BNT162b2	1(0-27)	Alpha (B.1.1.7)	188/6857	5828/119419	31 (20, 41)	119/4116	3997/42062	28 (13, 41)	..	..	..	..	..	..	..	..	..	low
							Delta (B.1.617.2)	317/6986	3672/117263	12 (0, 22)	203/388	2439/40504	18 (4, 31)	..	..	..	..	..	..	..	..	..	
						1(≥28)	Alpha (B.1.1.7)	273/14324	5828/119419	38 (29, 45)	182/2466	3997/42062	27 (13, 39)	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	163/14114	3672/117263	30 (17, 41)	92/2376	2439/40504	33 (15, 47)	..	..	..	..	..	..	..	..	..	
						2(0-13)	Alpha (B.1.1.7)	59/7277	3672/117263	73 (64, 79)	20/898	3997/42062	78 (65, 86)	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	15/7233	3672/117263	66 (43, 80)	5/883	2439/40504	84 (61, 93)	..	..	..	..	..	..	..	..	..	
						2(≥14)	Alpha (B.1.1.7)	104/53575	3672/117263	92 (90, 93)	34/4360	3997/42062	92 (88, 94)	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	208/53679	3672/117263	79 (75, 82)	75/4401	2439/40504	83 (78, 87)	..	..	..	..	..	..	..	..	..	
					AZD1222	1(0-27)	Alpha (B.1.1.7)	567/15137	5828/119419	9 (-1, 17)	398/4634	3997/42062	17 (6, 26)	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	293/14863	3672/117263	7 (-7, 19)	186/4422	2439/40504	23 (7, 36)	..	..	..	..	..	..	..	..	..	
						1(≥28)	Alpha (B.1.1.7)	956/51572	5828/119419	37 (32, 42)	591/10322	3997/42062	39 (32, 45)	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	776/51392	3672/117263	18 (9, 25)	511/10242	2439/40504	33 (23, 41)	..	..	..	..	..	..	..	..	..	
						2(0-13)	Alpha (B.1.1.7)	99/13818	3672/117263	64 (56, 71)	57/1774	3997/42062	65 (54, 74)	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	265/13975	3672/117263	25 (14, 35)	160/1877	2439/40504	37 (23, 48)	..	..	..	..	..	..	..	..	..	
						2(≥14)	Alpha (B.1.1.7)	100/32588	3672/117263	73 (66, 78)	36/1999	3997/42062	81 (72, 87)	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	231/32719	3672/117263	60 (53, 66)	126/2089	2439/40504	61 (51, 70)	..	..	..	..	..	..	..	..	..	
Alejandro Jara*	2021.7	cohort study	Chile	general population	CoronaVac	1(14-28)		20865/697883 52	185633/61486 8240	16 (14, 17)	..	..	..	3370/7069079 6	18034/620894 706	37 (35, 40)	1154/7083659 7	6359/6212264 31	45 (41, 48)	847/70854187	2786/6214264 77	46 (41, 50)	low

					2(≥14)		12286/91671797	185633/614868240	66 (65, 67)	..	..	..	1462/92445333	18034/620894706	88 (87, 88)	360/92622083	6359/6212264 31	90 (89, 91)	409/92514261	2786/6214264 77	86 (84, 88)		
Nabeel Khan	2021.5	cohort study	USA	Patients With Inflammatory Bowel Disease With Diverse Exposure to Immunosuppressive Medications	mRNA-1273+BNT162b2	1(≥14)		..	..	-1 (-50, 32)	..	..	..	..	..	..	..	..	9 (-114, 61)	..	..	-27 (-116, 25)	low
					2(≥7)		..	..	69 (44, 83)	..	..	..	..	..	..	..	..	49 (-36, 81)	..	..	57 (-19, 85)	
Skowronski, D. M.	2021.7	case control	British Columbia, Canada	adults 70 years and older	Not mentioned	1(0-13)	Alpha (B.1.1.7)	140/3218	200/4267	18 (-3, 34)	..	..	..	..	..	..	..	..	..	..	..	..	low
							Gamma (P.1)	81/3159	109/4176	11 (-19, 34)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	81/3159	109/4176	17 (-10, 38)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(14-20)	Alpha (B.1.1.7)	66/2097	200/4267	48 (30, 61)	..	..	..	..	..	..	..	..	..	..	..	..	
							Gamma (P.1)	53/2084	109/4176	24 (-19, 34)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	24/2055	126/4193	63 (43, 77)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(21-27)	Alpha (B.1.1.7)	52/2244	200/4267	55 (37, 67)	..	..	..	..	..	..	..	..	..	..	..	..	
							Gamma (P.1)	30/2222	109/4176	52 (26, 69)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	15/2207	126/4193	74 (55, 85)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(28-34)	Alpha (B.1.1.7)	22/1812	200/4267	76 (62, 85)	..	..	..	..	..	..	..	..	..	..	..	..	
							Gamma (P.1)	18/1808	109/4176	66 (42, 80)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	12/1802	126/4193	70 (43, 84)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(35-41)	Alpha (B.1.1.7)	11/1191	200/4267	81 (64, 90)	..	..	..	..	..	..	..	..	..	..	..	..	
							Gamma (P.1)	9/1189	109/4176	68 (35, 85)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	9/1189	126/4193	65 (27, 83)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥21)	Alpha (B.1.1.7)	103/6424	200/4267	67 (57, 75)	..	..	..	..	..	..	..	..	..	..	..	..	
							Gamma (P.1)	71/6392	109/4176	61 (45, 72)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	42/6363	126/4193	72 (58, 81)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥42)	Alpha (B.1.1.7)	18/1168	200/4267	67 (44, 81)	..	..	..	..	..	..	..	..	..	..	..	..	
							Gamma (P.1)	14/1164	109/4176	53 (12, 75)	..	..	..	..	..	..	..	..	..	..	..	..	
							any	6/1156	126/4193	72 (33, 88)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(0-13)	any	..	..	14 (0, 26)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(14-20)	any	..	..	43 (30, 53)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(35-41)	any	..	..	75 (63, 83)	..	..	..	..	..	..	..	..	..	..	..	..	
Pablo Knobel	2021.6	cohort study	Spain	Health Care Workers	mRNA-1273+BNT162b2	1(<14)		32/2076	94/6767	..	..	..	..	..	..	..	..	..	..	..	..	..	moderate
						1(≥14)		19/2350	94/6767	..	..	..	..	..	..	..	..	..	..	..	..	..	
						2(≥7)		7/5530	94/6767	..	..	..	..	..	..	..	..	..	..	..	..	..	
Jamie Lopez Bernal	2021.5	case control	UK	older adults	BNT162b2	1(0-3)		167/444	8988/24706	-22 (-48, 0)	..	..	..	..	..	..	..	..	..	..	..	..	low
						1(4-6)		179/420	8988/24706	-28 (-56, -5)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(7-9)		257/509	8988/24706	-48 (-77, -23)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(10-13)		284/645	8988/24706	-13 (-33, 4)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(14-20)		336/798	8988/24706	-6 (-23, 8)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(21-27)		118/406	8988/24706	36 (21, 49)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(28-30)		72/362	8988/24706	59 (46, 68)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(35-41)		65/339	8988/24706	51 (34, 63)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥42)		59/455	8988/24706	61 (45, 71)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(0-3)		45/161	8988/24706	41 (17, 59)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(4-6)		30/110	8988/24706	43 (12, 63)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(7-13)		28/229	8988/24706	79 (68, 86)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(≥14)		41/675	8988/24706	85 (79, 89)	..	..	..	..	..	..	..	..	..	..	..	..	
Shrotri, M.*	2021.6	cohort study	England	residents of long-term care facilities	BNT162b2	1(0-6)		11·95/10000	21·39/10000	16 (-81, 61)	..	..	..	..	..	..	..	..	..	..	..	..	low
						1(7-13)		8·13/10000	21·39/10000	-11 (-88, 35)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(14-20)		11·8/10000	21·39/10000	23 (-58, 63)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(21-27)		26·67/10000	21·39/10000	6 (-79, 50)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(28-34)		28·26/10000	21·39/10000	53 (-6, 80)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(35-48)		38·6/10000	21·39/10000	65 (29, 83)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥49)		26·24/10000	21·39/10000	62 (7, 85)	..	..	..	..	..	..	..	..	..	..	..	..	
					AZD1222	1(0-6)		19·79/10000	21·39/10000	51 (1, 100)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(7-13)		19·08/10000	21·39/10000	42 (4, 65)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(14-20)		25·1/10000	21·39/10000	5 (-84, 50)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(21-27)		16·32/10000	21·39/10000	27 (-44, 63)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(28-34)		8·51/10000	21·39/10000	67 (32, 84)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(35-48)		10·37/10000	21·39/10000	68 (34, 85)	..	..	..	..	..	..	..	..	..	..	..	..	
						1(≥49)		21·56/10000	21·39/10000	36 (-56, 74)	..	..	..	..	..	..	..	..	..	..	..	..	
Jamie Lopez Bernal	2021.7	case control	UK	general population	AZD1222	1(>21)	Alpha (B.1.1.7)	1776/44605	7313/103684	49 (45, 52)	..	..	..	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	1356/44185	4043/100414	30 (24, 35)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(≥14)	Alpha (B.1.1.7)	94/8338	7313/103684	75 (68, 79)	..	..	..	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	218/8462	4043/100414	67 (61, 72)	..	..	..	..	..	..	..	..	..	..	..	..	
					BNT162b2	1(>21)	Alpha (B.1.1.7)	450/9091	7313/103684	48 (42, 53)	..	..	..	..	..	..	..	..	..	..	..	..	low
							Delta (B.1.617.2)	137/8778	4043/100414	36 (23, 46)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(≥14)	Alpha (B.1.1.7)	49/15798	7313/103684	94 (92, 95)	..	..	..	..	..	..	..	..	..	..	..	..	low
							Delta (B.1.617.2)	122/15871	4043/100414	88 (85, 90)	..	..	..	..	..	..	..	..	..	..	..	..	
					BNT162b2+AZD1222	1(>21)	Alpha (B.1.1.7)	2226/53696	7313/103684	49 (46, 52)	..	..	..	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	1493/52963	4043/100414	31 (25, 36)	..	..	..	..	..	..	..	..	..	..	..	..	
						2(≥14)	Alpha (B.1.1.7)	143/24136	7313/103684	88 (85, 90)	..	..	..	..	..	..	..	..	..	..	..	..	
							Delta (B.1.617.2)	340/24333	4043/100414	80 (77, 82)	..	..	..	..	..	..	..	..	..	..	..	..	